Data Management in Galaxy

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The recent surge in life science data calls for adequate infrastructures and tools that facilitate FAIR (Findable, Accessible, Interoperable and Reusable) data access, manipulation, analysis and visualisation.

Galaxy is a well-known open-source platform that supplies tools and workflow to:

- access and collect data from reference databases, external repositories and other data sources,
- use tools from various domains that can be plugged into workflows,
- run code in interactive environments (RStudio, Jupyter...),
- capture the metadata of data analyses, thus ensuring their reproducibility,
- share and publish results, workflows and visualizations.

The **ELIXIR Research Data Management Kit (RDMkit)** provides tools, examples and best practice to researchers and data managers from various domains. Here, we present the Galaxy features mapped to the life cycle stages of the RDMkit.

SCAN ME

ACCOUNT CLEANING

✓Storage dashboard to manage quota
✓Bulk (permanent) delete
✓Quota temporarily extendable
✓Multiple quota per object storage (WIP)

IMPORT ARTEFACTS

✓Histories (own, shared by others)
✓Workflows from the WorkflowHub

ACCESS TO DATABASES

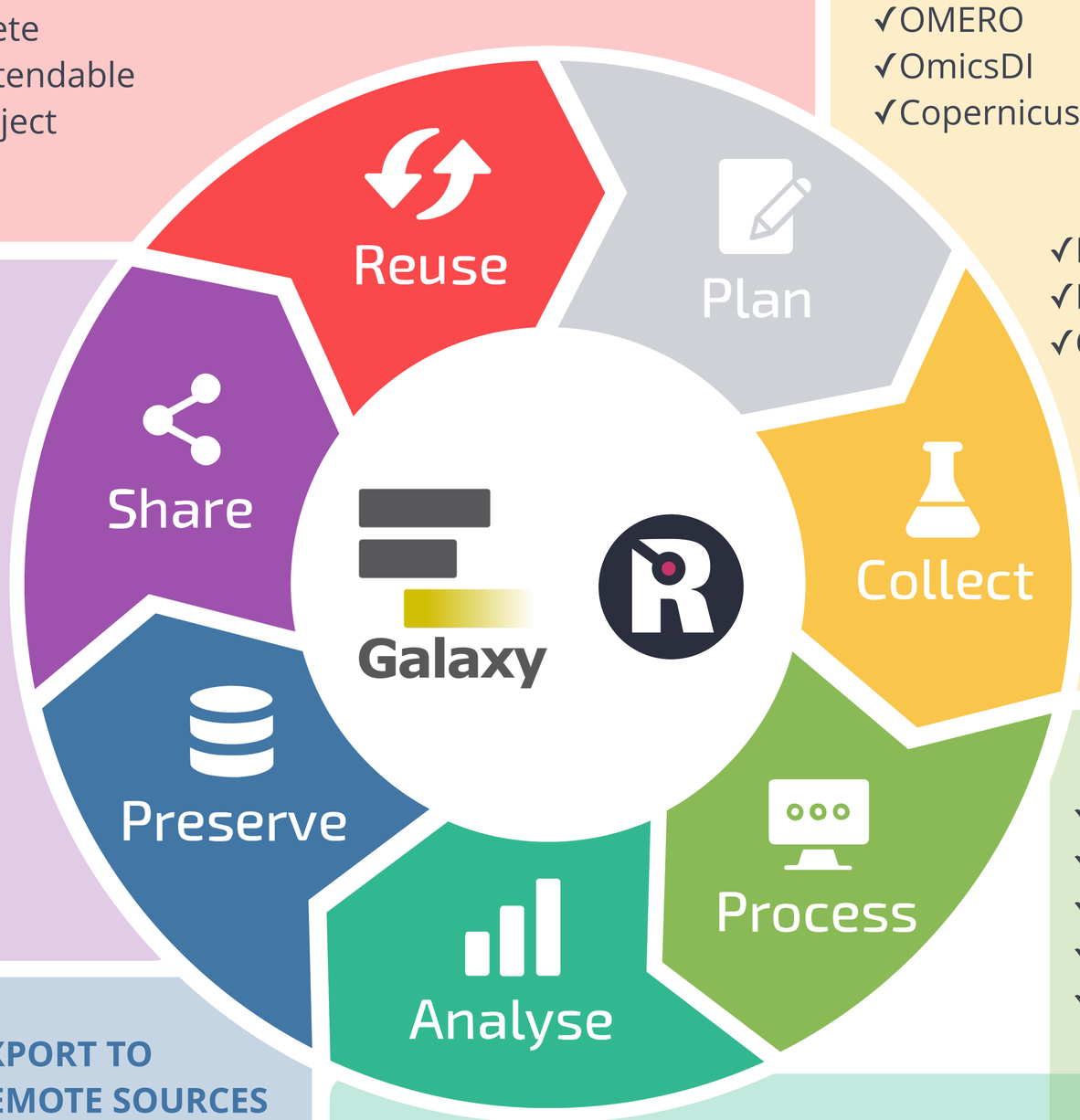
✓UniProt
✓InterMine
✓OMERO
✓OmicsDI
✓Copernicus
✓UCSC genome browser
✓NCBI datasets
✓INSDC (ENA)
✓PDB
✓3rd-party databases

CUSTOMISED DATA ACCESS

✓Data libraries
✓BYOD (Dropbox...)
✓On-demand reference data
✓*Deferred* data from remote locations

LIMS INTEGRATION

✓Connect to sequencing facilities
✓Rich API for integration with LIMS



SHARE ARTEFACTS

✓Datasets
✓Histories
✓Workflows
✓Visualizations
✓GA4GH Beacon (WIP)
✓DRS server

SHAREABILITY

✓RBAC (Role-Based Access Control)
✓ One user
✓ A group of users
✓ Public

DATA TRANSFORMATION

✓Data transformation tools
✓Quality Control
✓Data cleaning
✓Annotation
✓Interactive Tools (OpenRefine, RStudio, Jupyter Notebook)

EXPORT ARTEFACTS

✓Workflows
✓History
✓Datasets

FORMATS

✓Archive file
✓BioCompute Object
✓RO-Crate (WIP)

EXPORT TO REMOTE SOURCES

✓FTP
✓Dropbox
✓S3 Bucket
✓AWS
✓GDrive
✓Nextcloud
✓WebDav
✓Google Cloud Storage

2,900 DOMAIN-SPECIFIC TOOLS

✓Genomics
✓Cancer research
✓Computational chemistry
✓Climate science
✓Ecology
✓Imaging
✓Material science
✓Metabolomics
✓Microbiome
✓Machine Learning
✓Plant Biology
✓Proteomics
✓Single-cell omics

IMPORT WORKFLOWS

✓WorkflowHub
✓Dockstore
✓GA4GH TRS API

METADATA HANDLING

✓Provenance tracking
✓Automatic metadata enrichment
✓Bulk (meta)data manipulation

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